

Jinchen Zhao

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EDUCATION

Yale University Ph.D. in Applied Physics	Aug 2023 – May 2028 (expected) <i>New Haven, CT</i>
Duke Kunshan University & Duke University (dual degree) B.S. in Applied Mathematics & Computational Science	2019 – 2023 <i>Kunshan, China / Durham, NC</i>

RESEARCH INTERESTS

Quantum Information, Deep Learning, Quantum Optics. Quantum-Enhanced Physical Neural Networks, Nonlinear Photonics, Quantum Phase Transitions.

RESEARCH EXPERIENCE

Yale University Graduate Research Assistant (Advisor: Dr. Logan Wright)	Aug 2023 – Present <i>New Haven, CT</i>
· Studying unconventional computing platforms (optical, quantum, and thermodynamic), focusing on the role of analog noise in computation at ultra-low energy scales and pathways toward practical, scalable applications.	
Duke Kunshan University Undergraduate Research Assistant (Advisor: Dr. Myung-Joong Hwang)	2021 – 2023 <i>Kunshan, China</i>
· Discovered the frustration of photons during the superradiant phase transition of a Dicke lattice model and studied its novel critical behaviors.	
Duke University Undergraduate Research Assistant (Advisor: Dr. Kenneth Brown)	Spring 2022 <i>Durham, NC</i>
· Developed an auto-loading module for the trapped ion system, which tested successfully on the device and was added to the official control system.	
Duke Kunshan University Undergraduate Research Assistant (Advisor: Dr. habil. Kai Huang)	2019 – 2021 <i>Kunshan, China</i>
· Monitored projectile motion under opaque granular material by embedded inertial measurement unit, and studied granular impact under micro-gravity.	

PUBLICATIONS

- [1] **Jinchen Zhao** and Logan Wright, *Quantum-Enhanced Analog Computing*, in preparation.
- [2] Tivadar Pongó*, Tianhui Liao*, **Jinchen Zhao**, Valentin Dichtl, Simeon Völkel, Raúl Cruz Hidalgo, and Kai Huang, *Nature of granular drag in microgravity*, [arXiv:2603.03006](https://arxiv.org/abs/2603.03006), under review at *Phys. Rev. Lett.*
- [3] **Jinchen Zhao** and Myung-Joong Hwang, *Anomalous criticality with bounded fluctuations and long-range frustration induced by broken time-reversal symmetry*, *Phys. Rev. Research* **5**, L042016 (2023).

- [4] **Jinchen Zhao** and Myung-Joong Hwang, *Frustrated Superradiant Phase Transition*, [Phys. Rev. Lett. 128, 163601 \(2022\)](#).
- [5] Stefan Köstler, **Jinchen Zhao**, Chen Lyu, Simeon Völkel, and Kai Huang, *Embedded Inertial Sensor for Tracking Projectile Impact on Granular Media*, [EPJ Web Conf. 249, 15007 \(2021\)](#).

PRESENTATIONS

- [1] *Tensor Processing with Nonlinear Optical Media*, Frontiers in Optics, 2026.
- [2] *Anomalous Multicritical Phenomena and Frustration Induced by Synthetic Magnetic Fields*, APS March Meeting, 2023.

INDUSTRY EXPERIENCE

NTT Research, Physics & Informatics Lab Summer 2026
Research Intern *Remote*

- Studied the role of quantum optics in optical computing, and tensor computing in programmable nonlinear multimode waveguides.

World Economic Forum 2022
University-Corporation Lab

- Led a team of 5 students to map the East Asian quantum computing ecosystem, synthesizing policy, industry, and academic data for the WEF Technology and Quantum Computing Topic Leader.

HONORS

Anthropic Fellows Program, Offer (<100 of 10,000+ applicants)	2026
SPIE Optics and Photonics Scholarship (international scholarship for optics research)	2026
Jane Street Graduate Research Fellowship, Honorable Mention (<50 of 1,300+ applicants)	2026
Roberts Innovation Fund, Yale Engineering (seed award for research commercialization)	2025
Sterling Prize Fellowship, Yale University (top incoming graduate students)	2023
Zu Chongzhi Math Student First Award, Duke Kunshan University (top mathematics graduate)	2023
<i>Magna Cum Laude</i> , Duke Kunshan University	2023
Graduation with Distinction, Duke Kunshan University	2023

SERVICE

Advisor of 5 undergraduate research projects, Yale University
Reviewer, *Nature Photonics*
Reviewer, *Nature Communications*
Reviewer, *Light: Science & Applications*
Reviewer, *Journal of the Optical Society of America B*
Teaching Fellow, APHY 5750 *Physics of AI*, Yale University
Teaching Fellow, PHYS 180 *University Physics*, Yale University

SKILLS

Programming: Python (PyTorch, QuTiP, Qiskit), Verilog/HDL

Hardware: RFSoc/FPGA (Vivado), RF & photonic instrumentation control

Tools: Mathematica, Git, L^AT_EX